Generic Event Boundary Candidates for Lecture Segmentation

A Pilot Study of Coverage, Density, and Stopping Cost

Hyojung Ryu

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Abstract

We examine whether a pretrained RGB generic event boundary detector provides useful candidate timestamps for reviewed lecture segmentation, and how adaptive stopping (D) changes observed cost relative to full 32-step inference (F32). Two development and six evaluation clips were prepared. Evaluation labels originated from AI drafts reviewed by one user without viewing GEBD or count-matched uniform candidates; this is not independent free annotation. The frozen evaluation contains 17 primary Focus (L1) and 21 internal (L2) boundaries. Four L2 boundaries retain the same Function label, and two child Focus descriptions differ from the parent, so L2 is an exploratory evaluation of reviewed internal boundaries. At ±3 s, F32 covers 14/17 L1 and 12/21 L2 boundaries; D covers 15/17 and 13/21. Count-matched uniform controls U-F32/U-D cover 9/17 and 10/17 L1, and 14/21 and 13/21 L2 boundaries. Density differs sharply by clip. Summing video medians from three repetitions without a feature cache, D reduces the measured model interval by 53.5% and the decoding-inclusive pipeline by 35.9%; warmup, accuracy runs, model loading, and disk writes are excluded. This pilot supports examining candidate utility by level, density, and measured cost, not general accuracy preservation or superiority over fixed low-step inference.

1 Introduction

Lecture videos contain changes of topic and instructional activity that need not coincide with visible event boundaries. We investigate whether existing visual event boundaries can supply timestamps for subsequent semantic review. Candidate coverage is only a prerequisite for that use: a large candidate set can obtain high recall while creating a substantial review burden. It neither labels the activity nor establishes useful final segmentation or retrieval.

Generic event boundary detection (GEBD) addresses taxonomy-free event transitions [1]. DiffGEBD generates boundaries by denoising conditioned on visual features [2]. A prior project manuscript studies boundary stability as a signal for stopping this computation [3]. Here we reuse its implementation without retraining and ask how the resulting candidates behave under reviewed lecture labels. We evaluate both full computation and stopping on the same 32-step trajectory, rather than propose a new semantic or multimodal model.

Our intended hierarchy distinguishes primary Focus intervals (L1) from Function changes within them (L2). The actual reviewed labels do not fully instantiate that definition: four internal boundaries have no Function-label change, and two child Focus descriptions disagree with their parent. We retain the frozen labels and report L2 as exploratory coverage of reviewed internal segmentation. This qualification changes the interpretation of the task, not its post hoc scores.

Related work and the comparison gap

GEBD annotation considers a dominant subject and a finer temporal granularity, with subjective boundary judgments [1, §3]. DiffGEBD uses temporal self-similarity and a diffusion decoder [2, §4]; we treat its output as candidates rather than final semantic decisions. VidChapters-7M evaluates chapter boundaries, including second-based tolerances [4, §4]; therefore neither timestamp recall nor chapter segmentation is new here.

Lecture topic segmentation models multimodal coherence [5, §3]. Sapena and Onaindia classify lecture segments using a hierarchy of teaching activities such as Theory, Exercise, and Example [6, §3]. The activity categories overlap our motivation, but that class hierarchy is not the temporal nesting of Focus intervals and activities within them. Our contribution is a small, frozen-checkpoint diagnostic of candidate location, count, and observed stopping cost, with an explicit audit of the reviewed labels. We do not establish novelty through the names Focus and Function.

Questions and scope

We ask: (i) how does coverage differ between L1 and reviewed L2 internal boundaries relative to equal-count uniform timestamps; (ii) how do density, tolerance, and placement phase affect this comparison; and (iii) what model and pipeline costs are observed for stopping relative to F32? The available evidence is six convenience-sampled evaluation clips, AI-assisted review by one user, and a single frozen model. There is no final independent test, fixed low-step comparison, semantic-model ranking, or retrieval experiment.

2 Data Labels and Frozen Protocol

2.1 Data and annotation provenance

We use eight Korean lecture/tutorial clips of approximately eight minutes each: two development clips and six evaluation clips (about 64 minutes total). Source videos are distinct, and the recorded speaker/series groups are disjoint across the development/evaluation split. Availability and language guided selection before detector outputs were used. We used the two development clips to establish boundary annotation guidelines and a review procedure before applying that procedure to the evaluation clips. Identifiers such as eval01 and eval03b denote individual evaluation clips in the tables and figures.

The original plan called for independent free evaluation annotation. In practice, one user reviewed and edited AI proposals based on existing ASR and sampled frames while watching the videos. The evaluation contains 23 L1 and 44 L2 segments, yielding 17 L1 and 21 internal boundaries. The user confirmed viewing only AI drafts and videos during review. A review record incorrectly marked detector predictions as viewed; we corrected it against the original records and change history. The restricted review interface also prevented access to GEBD and uniform candidates. These records support non-exposure to candidate predictions, but they do not make the labels independently authored, demonstrate inter-rater agreement, or remove AI anchoring.

The annotation guideline defines L1 as a maximal interval of primary Focus and intended L2 as a maximal interval of primary Function within its parent. In eval01, however, the boundaries at 268.02, 308.00, 398.00, and 445.00 s are explanation→explanation. After the last two, child descriptions concern AI and numerical/data-analysis ecosystems while the parent description remains web development. Parent IDs and temporal coverage passed structural checks; semantic consistency did not follow. We keep all four boundaries and the parent descriptions unchanged. L2 results are exploratory coverage of reviewed internal boundaries, not pure Function-transition performance.

Times are relative to clip start. We exclude clip endpoints, the first L1 start, and each parent's first L2 start; L1 boundaries are not duplicated in internal L2. Low-confidence boundaries remain in the main evaluation. The labels used in analysis were finalized after user review and frozen before candidate inspection. No labels were revised after detector predictions were viewed.

2.2 Window inference and adaptive stopping

We reuse the official pretrained DiffGEBD checkpoint [2] and the inference and stopping implementation used in the prior project manuscript [3]. Each clip produces 95 windows of length 10 s and stride 5 s, with an end-aligned tail where needed. Each window supplies 100 integer-linearly sampled native frames resized to 224×224. Frame timestamps map predictions to clip seconds. This is windowed transfer to long lectures, not the original short-video evaluation.

F32 runs all 32 steps. D returns when consecutive decoded sets have equal cardinality and maximum sorted-pair displacement ≤3 native frames (h=1); both-empty sets are stable. The earliest stop is step 2, with fallback at 32. The tolerance is 3/FPS seconds on the CFR derivative, not the original benchmark's annotation-frame unit or the 100-sample index. The adaptive unit is a window. Detection threshold 0.5, CFG conditional+7×(conditional−unconditional), η=0, FP32, and batch 1 (AMP/TF32 off) are shared. Stable ID-to-seed mapping and actual initial-noise hashes agree for all 570 paired windows.

Raw candidates are sorted into clusters of complete span ≤0.5 s, represented by the median; no forced splice boundary is added or removed. For each video and base method, uniform controls use its final merged count N. The jth uniform candidate is at $(j+0.5)L/N$ for $j=0,\ldots,N-1$, where L is the clip duration; N=0 gives an empty set. U-F32/U-D control location at the observed count. They receive N from the model and do not predict a deployable candidate density. Fixed offsets 0.25 and 0.75 are auxiliary checks.

2.3 Metrics and aggregation

Candidate Recall@δ is the fraction of reference boundaries within δ seconds of any candidate. We report ±3 s primarily and ±6 s secondarily. One-to-one recall and boundary P/R/F1 use maximum-cardinality matching, then minimum total absolute error and deterministic ties. Pooled results sum covered boundaries or TP/FP/FN; they are not means of video F1. Zero-reference recall/F1 and zero-candidate precision are NA. Positive-reference empty predictions have recall/F1 zero. Density is merged candidates per minute; L1, L2 internal, and their union remain separate.

2.4 Actual cost measurement

On one RTX 4060 Ti, we measure each video/method three times, alternating order after warmup. Model time spans synchronized GPU-input readiness through backbone, encoder, denoising, stopping, and CPU boundary extraction. Pipeline time additionally includes independent PTS scanning, decoding, preprocessing, transfer, coordinate mapping, and merging. No feature cache is used; OS disk cache is uncontrolled. Warmup, separate accuracy/noise checks, model loading, and

result writes are excluded. We sum per-video medians, not three repetitions. All 36 timed runs match the saved accuracy outputs and call counts. Uniform supplied-N generation has no backbone/decoder cost; its separate generation time is not comparable to model inference.

3 Results Coverage Density and Observed Cost

Table 1. Six evaluation clips; 17 L1 and 21 reviewed internal L2 boundaries. CR: Candidate Recall; 1:1 R: one-to-one recall. AI-assisted drafts reviewed by one user; GEBD/uniform candidates not viewed.

A. Coverage within ±3 s and candidate density

Method	L1 CR / 1:1 R (%)	L2 CR / 1:1 R (%)	Candidates	Candidates/min
F32	82.4 / 82.4	57.1 / 57.1	772	16.08
D	88.2 / 88.2	61.9 / 61.9	753	15.69
U-F32	52.9 / 52.9	66.7 / 66.7	772	16.08
U-D	58.8 / 58.8	61.9 / 61.9	753	15.69

B. Measured cost: sums of six per-video medians (three repetitions each)

Method	Model (s)	Pipeline (s)	Mean steps/window	Decoder calls/pass
F32	306.00	468.87	32.000	36,480
D	142.21	300.36	2.074	2,364
U-F32	—	—	0.000	0
U-D	—	—	0.000	0

Note. All low-confidence boundaries are retained. Four L2 boundaries have unchanged Function labels; two child Focus descriptions differ from their parent. U-F32/U-D receive each base method's video-specific merged N (offset 0.5); they do not predict density. Cost excludes warmup, accuracy checks, model loading, and writes; no feature cache. Model time includes backbone through CPU boundaries; pipeline adds decoding, preprocessing, transfer, and merging. Mean-step denominator: 570 windows. Uniform model/pipeline cost: NA.

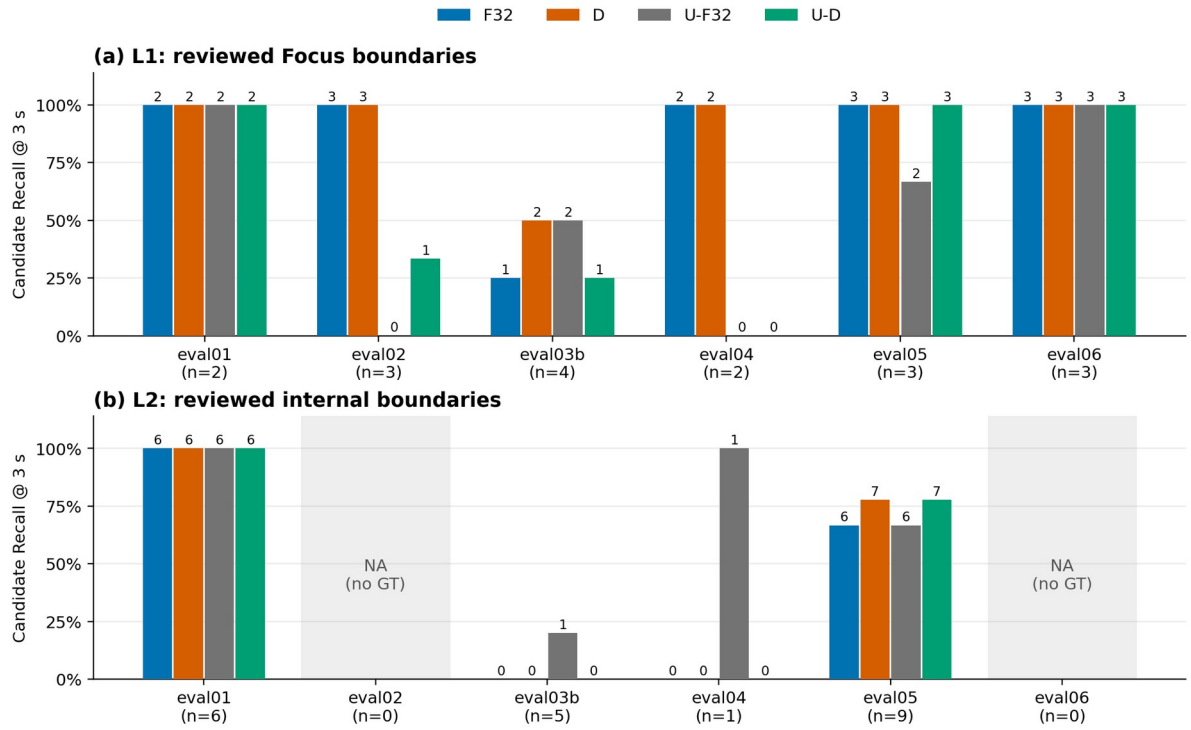


Figure 1. Coverage by video at ±3 s. Bar labels are covered reference counts; n is the level's reference count. The F32/D merged candidate counts (N) by clip are eval01 351/339, eval02 24/19, eval03b 14/7, eval04 9/2, eval05 62/61, and eval06 312/325; the corresponding uniform controls receive those counts. L2 with no reference boundaries is NA. AI-assisted labels were reviewed by one user without GEBD/uniform candidates. L2 is exploratory internal-boundary coverage, not pure Function accuracy.

With AI-assisted labels reviewed by one user, F32/D exceed their equal-count uniform controls on L1; D also covers one more boundary per level than F32 (Table 1). L2 retains four same-Function splits and two child/parent Focus inconsistencies; it is exploratory internal-boundary coverage, not pure Function performance. U-F32 exceeds F32 on L2 and U-D equals D. Changing both count and location, F32→D does not isolate location quality or guarantee accuracy.

The video-specific counts reported with Figure 1 show that eval01/eval06 contribute 663/772 F32 candidates (85.88%), versus 14 in eval03b and 9 in eval04. Dense clips saturate for uniform controls too: spacing ≤6 s guarantees ±3 s coverage.

L2 with no reference boundaries remains NA. Observed model/pipeline costs fall 53.5%/35.9% (Table 1), using cache-free processing scopes and excluding warmup, accuracy checks, loading, and writes (§2.4). Steps average over 570 windows. No fixed low-step efficiency ranking follows.

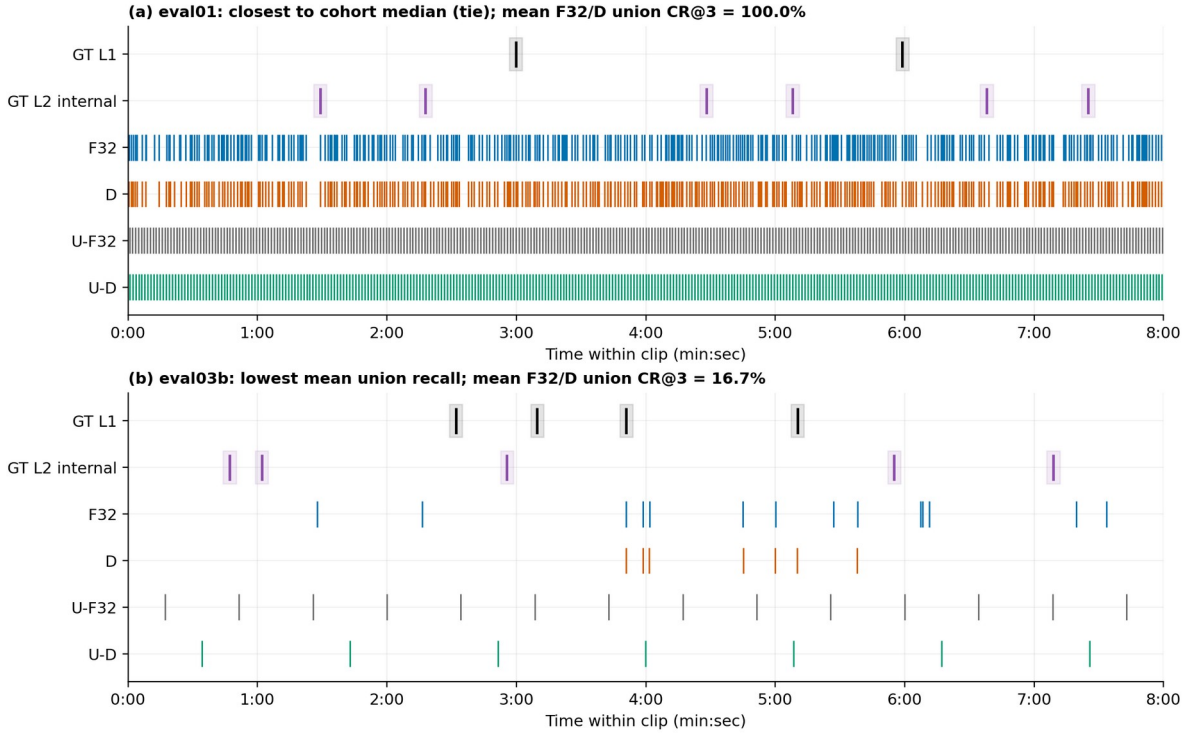


Figure 2. High-density coverage saturation (eval01) and low candidate coverage (eval03b). Ticks share clip-relative time; shading is ±3 s. Exploratory selection: nearest median mean F32/D union coverage (ties by ID) and lowest distinct clip. Uniform controls receive each base N. AI-assisted, one-user-reviewed labels were not independently authored; GEBD/uniform candidates were not viewed. L2 retains four unchanged-Function splits and two child/parent Focus inconsistencies. These are candidate timelines, not final semantic decisions.

4 Diagnostics Limitations and Conclusion

Tolerance and placement matter. At ±6 s, F32 and D each cover 13/21 internal boundaries; U-F32/U-D cover 19/21 and 16/21. Across offsets 0.25/0.5/0.75, uniform L2 coverage at ±3 s ranges from 12–14 (U-F32) and 11–15 (U-D). No favorable phase is selected. Figure 2 shows both dense saturation and low coverage, rather than only GEBD-favorable cases.

Sampled frames plus existing ASR support static-slide misses at eval03b's 151.98 s L1 and 47.08 s L2 boundaries; both labels have low confidence. At eval05's 441.12 s internal boundary, F32's nearest candidate is 4.053 s early, recovered only at ±6 s. For the qualitative diagnostic, we selected up to the first two cases per level and error type under a rule set before inspection, yielding nine cases in total. These are not new independent human error judgments or direct audio listening.

Removing candidates within the fixed ±0.5 s internal-splice neighborhood reduces L1 coverage from 14 → 12 (F32) and 15 → 12 (D), while L2 coverage is unchanged. This diagnostic does not replace the main results; after filtering, uniform/base counts no longer match. Raw-to-median merging changes none of the 152 reference-coverage checks across methods, levels, and tolerances. Splice proximity does not establish a technical cause, and no alternative-window causal ablation was run.

Limitations include one Korean domain, six convenience clips, AI anchoring, one reviewer, low-confidence times, L2/parent-Focus inconsistencies, windowed inference, and the changed stopping-frame unit. No labels were revised after seeing predictions. Semantic, shot, retrieval, and fixed low-step comparisons remain outside this pilot.

Conclusion

These AI-assisted, single-user-reviewed labels were not independently authored; GEBD/uniform candidates were not viewed. Coverage differs by level and density under equal-count uniform controls. L2 remains exploratory: four unchanged-Function splits and two parent-Focus inconsistencies preclude pure Function claims. D's cost is lower than F32 for the measured cache-free model interval (backbone through CPU boundaries) and pipeline (also decoding/preprocessing/transfer/merging), excluding warmup, accuracy checks, loading, and writes. This establishes neither general accuracy preservation nor superiority over fixed low-step methods. Any future semantic adjudication needs a new label version; current labels and main results remain frozen.

References

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